



SILVER OAK UNIVERSITY
School of Technology, Design And Computer Application
Silver Oak College of Computer Application
Master of Science Data Science
Course Name: Data Analytics Using Tools-Practical
Course Code: DSM5042C
Semester: 2nd

Prerequisite: Fundamental knowledge of data management, statistical concepts, and prior experience with spreadsheets, basic data visualization techniques, and familiarity with analytical reasoning and problem-solving approaches.

Course Objective: To develop students' competencies in data analytics by enabling them to acquire, process, explore, visualize, and interpret data, implement statistical and analytical techniques, design and apply predictive and prescriptive analytics workflows, and generate actionable insights for informed, data-driven decision-making in academic and professional contexts.

Teaching Scheme:

Teaching Scheme				
L	T	P	Contact Hours	Credit
0	0	4	4	2

List of Practicals:

Total Hours: 56

Sr. No.	Practical Name
1	A setup comprising software and tools used for acquiring, processing, and analyzing datasets to extract meaningful insights.
2	An online platform for writing, executing, and sharing code for data analysis.
3	Loading structured data stored in CSV format into analytical software for processing.
4	Reading Excel files to examine dataset schema, headers, and prepare for analysis.
5	Establishing a link between an analytical tool and a database to access data.
6	Classification of data into numerical, categorical, or textual forms for appropriate analysis.
7	Summarization of data using measures like mean, median, and mode.
8	Identifying and imputing absent data to ensure dataset completeness.
9	Identification of extreme values that may distort statistical analysis.
10	Ensuring each entry in a dataset is unique and consistent.
11	Maintaining uniform representation and formatting across datasets.
12	Scaling numerical data into a specific range for comparative analysis.
13	Transforming data to have a mean of zero and standard deviation of one.

14	Converting categorical data into numerical representations.
15	Representing categorical variables as binary vectors to prevent ordinal bias.
16	Deriving new attributes from existing data to enhance analysis.
17	Identifying relevant variables to improve model performance.
18	Examining a single numerical variable to understand distribution and characteristics.
19	Studying frequency distributions of a single categorical variable.
20	Evaluating relationships between two variables using statistical or visual methods.
21	Examining relationships between categorical variables using contingency tables.
22	Studying interactions among three or more variables simultaneously.
23	Measuring the strength and direction of relationships between variables.
24	Summarizing data using mean, median, and mode.
25	Quantifying the spread of data using variance, standard deviation, and range.
26	Describing the likelihood of different outcomes in a dataset.
27	Comparing the means of two groups to assess statistical significance.
28	Comparing means among three or more groups to determine differences.
29	Evaluating independence between categorical variables.
30	Modeling the relationship between one independent and one dependent variable.
31	Modeling the relationship between multiple independent variables and a dependent variable.
32	Representing the relationship between variables graphically.
33	Machine learning using labeled data to predict outcomes.
34	Predicting probability of a binary outcome based on independent variables.
35	Predicting outcomes by splitting data into branches based on feature values.
36	Summarizing model performance using counts of true positives, false positives, true negatives, and false negatives.
37	Proportion of correct predictions in a classification model.
38	Proportion of true positive predictions among all positive predictions.
39	Proportion of true positive predictions among actual positive instances.
40	Harmonic mean of precision and recall for classification evaluation.
41	Graphical representation of the true positive rate versus the false positive rate.
42	Technique to assess model generalizability by testing on multiple subsets.
43	Identifying patterns in unlabeled data without predefined outcomes.
44	Partitioning data into groups based on similarity of features.
45	Graphically representing clusters to interpret data grouping.
46	Creating nested clusters based on similarity measures.
47	Discovering relationships and patterns between variables in datasets.
48	Identifying items that appear together regularly in transactional data.

49	Deriving actionable insights from identified patterns.
50	Applying data analysis techniques to improve decision-making in retail.
51	Dividing customers into groups for targeted strategies.
52	Using historical data to forecast future outcomes.
53	Automating repetitive data processing and analysis tasks.
54	Integrating data acquisition, processing, analysis, and visualization into a single workflow.
55	Presenting data visually using charts and interactive dashboards.
56	Estimating customer attrition and discovering co-purchase patterns for actionable insights.

Course Outcome:

Sr.No.	CO-Statement	Practical No.
CO-1	Identify fundamental concepts of data analytics, types of analytics, data types, CRISP-DM methodology, and ethical governance	1 to 7
CO-2	Perform data processing, cleaning, transformation, feature engineering, and exploratory data analysis	8 to 23
CO-3	Use statistical measures, correlation, regression, and visualization techniques to derive insights from data	24 to 32
CO-4	Integrate advanced analytics workflows, predictive and unsupervised modeling, and industry-oriented case studies to Analyze	33 to 56

Teaching & Learning Methodology:

1. Lecture-Based Learning
2. Hands-On Labs and Practical Sessions
3. Project-Based Learning
4. Group Activities and Collaborative Learning
5. Active Learning Strategies

Major Equipment:

1. Computers/Laptops with internet connectivity
2. Projector and display system for demonstrations

Books Recommended:

1. *Data Science for Business* by Foster Provost and Tom Fawcett
2. *Practical Statistics for Data Scientists* by Peter Bruce, Andrew Bruce, and Peter Gedeck

List of Open Source Software/learning website:

